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Roll No. .....

328356(28)

B. E. (Third Semester) Examination, April-May, 2016

(New Scheme)

(Et & T Branch)

DIGITAL LOGIC DESIGN

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : First part of each question is compulsory. Attempt any two parts from rest parts of the questions.

Unit-I

1. (a) (i) Represent CSVTUI'n ASCII & EBCDIC code.

(ii) Write all the laws of boolean algebra. 2

(b) Convert the following :

(i) $(1231)_{10} = (\quad)_2$

(ii) $(101110)_2 = (\quad)_{10}$

(iii) $(1998)_{10} = (\quad)_2$

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(iv) $(27)_{10} = (\quad)_2$

(v) $(4096)_{10} = (\quad)_2$

(vi) $(1011)_2 = (\quad)_8$

(vii) $(87)_{10} = (\quad)_8$ 7

(c) Assume that the data has been encoded in a 7-bit even parity Hamming code and the number 1011011 is received. Find out the bit in error. What will the corrected code be? 7

(d) (i) Find Gray code equivalent of the Hexaminal number $(A2C)_{16}$

(ii) Find XS-3 code for octal number $(267)_8$.

(iii) Find 5421 BCD equivalent of 83.

(iv) Construct the even parity seven bit hamming code for a word 1011. 7

Unit-II

2. (a) Express the Boolean function $F = A + \overline{B}C$ in SOP and POS form. 2

(b) Implement the following Boolean function using PAL with four inputs and 3 wide AND-OR structure. Also write the PAL programming table : 7

$$F_1(A, B, C, D) = \sum m(2, 12, 13)$$

$$F_2(A, B, C, D) = \sum m(7, 8, 9, 10, 11, 12, 13, 14, 15)$$

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$F_3(A, B, C, D) = \sum m(0, 2, 3, 4, 5, 6, 7, 8, 10, 11, 15)$

$F_4(A, B, C, D) = \sum m(1, 2, 8, 12, 13)$

(c) Simplify the following boolean expression using quine–Mc-Cluskey method in SOP form :

$F(A, B, C, D) = \pi M(2, 7, 8, 9, 10, 12)$ 7

(d) Minimize using K-map and implement using logic gates (AOI logic)

$$f = \sum_m (0, 2, 3, 8, 10, 12) + d(4, 6, 7)$$ 7

Unit-III

3. (a) Design 4 to 16 decoder using two 3 to 8 decoder. 2

(b) Design a full subtractor circuit using 2 input EX-OR gates and basic gates. Also give block diagram of full subtractor using half subtractor. 7

(c) (i) Design a 2-bit comparator circuit.

(ii) Design a BCD comparator circuit. 7

(d) (i) Implement the following :

$F(A, B, C) = \sum m(1, 3, 5, 6)$ with 4×1 multiplexer. Also design it using 8×1 multiplexer

(ii) Implement full adder using multiplexer.

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(iii) Implement the full subtractor using a 1 : 8 de-multiplexer. 7

Unit-IV

4. (a) Convert SR flip flop into JK flip-flop. 2

(b) What do you mean by race around condition in JK flip-flop. How it can be avoided in master slave flip-flop? 7

(c) Draw and describe the working of a parallel in parallel out. PIPO shift register. Explain how a number can be shifted in and out from such register. 7

(d) Draw and explain 4-bit Johnson ring counter. 7

Unit-V

5. (a) What is tristate logic? 2

(b) Explain and draw the circuit diagram for the following :

(i) Position logic RTL NOR gate

(ii) Positive logic DTL NAND gate

(iii) I^2L invert and I^2L NAND gate. 7

(c) Give the various characteristics of digital IC's? Basic of which the performance of IC's can be compared? 7

(d) What is ECL logic? Why it is popular? Give advantages and disadvantages of it. 7

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